DATA SCIENCE MINOR PROJECT REPORT

DATA-DRIVEN PREDICTION OF BANK TERM DEPOSIT SUBSCRIPTION USING MACHINE LEARNING

Project Semester: 5th

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DECLARATION

I, [Pawan Kumar Yadav], student of [Lovely School Of Computer Science] under the CSE/IT Discipline at Lovely Professional University, Punjab, declare that the project titled “Data-Driven Prediction of Bank Term Deposit Subscription Using Machine Learning” represents my project work, analysis and study. External data sources, software libraries, documentation and other resources used in the final submission will be acknowledged in the references.

I will verify that the reported numerical results match the analysis code executed for the submitted version. I will also verify the dataset source and complete all personal, faculty, project and repository link fields before submission.

Date: 29th October 2026

Registration No.: 12412363

Student Signature: Pawan Kumar Yadav

CERTIFICATE

This is to certify that Pawan Kumar Yadav, bearing Registration No. 12412363, has completed the project titled “Data-Driven Prediction of Bank Term Deposit Subscription Using Machine Learning” under my guidance and supervision as part of Predictive Analytics. To the best of my knowledge, the final submitted work reflects the student's project effort, subject to review of the final report and supporting files.

Supervisor Name and Signature: _____

Designation: _____

School: _____

Lovely Professional University, Phagwara, Punjab

Date: _____

ACKNOWLEDGEMENT

I express my gratitude to my faculty coordinator and the department for their guidance during this data-science minor project. Their advice on problem definition, exploratory analysis, implementation and presentation helped shape the work.

I acknowledge the provider of the dataset supplied for this analysis and the maintainers of the open-source Python tools used in data preparation, visualization and model evaluation. The exact original dataset source must be verified and cited by the student before submission.

This project provided an opportunity to practise the complete workflow: inspecting data, describing patterns, defining a prediction time, fitting alternative models, evaluating test-set performance and communicating limitations.

ABSTRACT

Telephone marketing campaigns compete for limited outreach time. This project examines whether historical bank-campaign records can support more informed prioritization of term-deposit outreach. The supplied CSV contains 11,162 rows and 17 columns. The target has 5,289 “yes” records (47.38%) and 5,873 “no” records (52.62%); the initial audit found no blank cells or exact duplicate rows.

Exploratory analysis describes customer attributes, financial fields, campaign timing and prior outcomes. Logistic regression and random forest are trained on a stratified 80/20 split. Numeric features are median-imputed and standardized; categorical fields are mode-imputed and one-hot encoded. Call duration is excluded because it is known only after a call and would leak post-contact information into a pre-call prediction.

On the held-out 2,233-row test set, random forest achieved 73.04% accuracy, 0.689 F1 and 0.787 ROC AUC; logistic regression achieved 69.55% accuracy, 0.636 F1 and 0.758 ROC AUC. Results support further study of a human-reviewed prioritization aid, not an automatic customer decision or a claim of causal campaign lift.

Keywords: bank marketing, term deposit, classification, exploratory data analysis, logistic regression, random forest, leakage control.

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Abbreviations: EDA — exploratory data analysis; F1 — harmonic mean of precision and recall; FN — false negative; FP — false positive; LR — logistic regression; ML — machine learning; RF — random forest; ROC — receiver operating characteristic; TN — true negative; TP — true positive; UCI — University of California, Irvine.

1. INTRODUCTION

Banks use direct outreach to explain products such as term deposits. A campaign may contain many contact opportunities, while staff time and customer attention are limited. Historical records can be analysed to identify patterns associated with prior subscriptions and to build a ranking that helps a human team organize outreach.

This project is a supervised binary-classification study. Each row contains customer-profile and campaign-history fields together with a deposit outcome marked yes or no. The intended use is to estimate subscription likelihood before a future call, using only information available at that time.

A predictive score is not an explanation of customer motivation. Observed historical patterns may reflect campaign selection, timing, or other unrecorded factors. The project therefore combines a prediction exercise with explicit leakage controls, error analysis and responsible-use limitations.

1.1 SOCIETAL CHALLENGE AND RELEVANCE

The societal challenge addressed is the inefficient allocation of outbound communication effort. When campaign teams contact records without a clear prioritization method, staff capacity may be spent on low-response cases, while a relevant offer may not reach a customer who would value it.

A data-informed ranking could help a team plan the order of optional calls and evaluate whether a campaign is reaching people who find the product relevant. This benefit remains a hypothesis until tested against an appropriate control group. A model that predicts historical responses does not by itself prove that model-ranked calls increase subscriptions or improve customer well-being.

The project does not automate product eligibility, credit, pricing or access decisions. Any practical deployment would need consent and contact-preference controls, documented purpose, human oversight, privacy protections, and compliance review.

1.2 PROBLEM STATEMENT

The project problem is to analyse bank campaign records and build a reproducible classifier that estimates whether a record is associated with a term-deposit subscription. The model must be evaluated on held-out examples and compared with a simple baseline.

Central question: How well can the supplied customer and campaign history fields distinguish records labelled deposit=yes from those labelled deposit=no, when post-call information is excluded?

The outcome is measured with precision, recall, F1, accuracy, balanced accuracy and ROC AUC. Model selection is based on held-out metrics, not on an assumption that a more complex method must be better. The chosen split is random and stratified; it is a first evaluation, not a future-period deployment test.

1.3 OBJECTIVES AND DELIVERABLES

Objective	Measurable completion criterion
Audit the supplied file	Report shape, target mix, missingness, duplicate count and ranges.
Describe data patterns	Produce labeled, readable EDA figures with subgroup sizes.
Control prediction time	Remove duration from all pre-call model inputs.
Compare classifiers	Evaluate logistic regression and random forest on the same split.
Communicate trade-offs	Report class-specific errors, metrics, limitations and human-review safeguards.

Expected deliverables

- A complete report containing all analysis, diagrams, charts, results and references.
- Separate Python source files for model execution and conceptual diagrams.
- Blank spaces for dataset/project/GitHub/LinkedIn URLs and student particulars.

1.4 SCOPE, RESEARCH QUESTIONS AND ASSUMPTIONS

Scope: one student-supplied CSV, one binary target, descriptive analysis, a stratified 80/20 train/test partition, a logistic regression baseline, and a random-forest comparison. The report does not identify individual customers because the provided file has no customer identifier.

Research questions

- What is the yes/no target mix and what quality issues are visible?
- How do observed outcomes vary across selected profile and campaign fields?
- Does the random forest improve the chosen test metrics over logistic regression?
- What residual errors and governance requirements remain?

Assumptions

The supplied labels yes/no are treated as the outcome for the recorded campaign. Categories such as unknown are retained as explicit values. The reported data unit is a row, not a verified unique customer. Original data provenance, licence and sampling details require confirmation by the student.

2. SOURCE OF DATASET

The project analyses the CSV supplied with the assignment: bank_1793283222379.csv. It contains 11,162 rows and 17 columns, including customer profile, campaign history, contact timing and the deposit target. The source file contains no embedded download URL, provider name, licence, or provenance metadata.

The field names and task resemble the bank telemarketing term deposit classification problem documented by the UCI Machine Learning Repository [1]. However, the supplied file contains 11,162 rows, while UCI describes 45,211 rows for bank-full.csv and 4,521 rows for bank.csv. This report therefore does not claim that the uploaded file is an exact copy of an official UCI release.

Dataset source link for final submission: [PASTE THE VERIFIED ORIGINAL DATASET URL HERE]

Source verification required

Before submission, confirm where this exact CSV was downloaded, whether its variables and units are documented, and whether its terms allow academic redistribution. Do not replace the blank dataset URL with an assumed link.

2.1 DATASET STRUCTURE

The file contains 17 fields: 16 candidate predictors and one supervised target. Numeric fields include age, balance, day, duration, campaign, pdays and previous. Categorical fields include job, marital status, education, default, housing, loan, contact, month and poutcome. The target field deposit contains yes/no.

The data are stored as individual records without a customer ID, so the exact observational unit and repeat-contact grouping cannot be independently verified. The target is the observed campaign outcome, not a future balance, income or lifetime value.

Pre-call prediction requires defining when each feature becomes available. The duration of the current call is measured only after the call and is excluded. Day and month are retained for this offline comparison but should be reviewed for availability in the intended future campaign.

2.2 DATA DICTIONARY: CLIENT PROFILE

Table 2 — Client-profile fields in the supplied CSV.

Field	Type	Description / observed values
age	Integer	Age in years; observed range 18–95.
job	Categorical	Occupation group; includes unknown.
marital	Categorical	Married, single or divorced.
education	Categorical	Primary, secondary, tertiary or unknown.
default	Category	Recorded credit-default status: yes/no.
balance	Integer	Recorded account balance; −6,847 to 81,204. Units not independently verified.
housing	Category	Recorded housing-loan status: yes/no.
loan	Category	Recorded personal-loan status: yes/no.

These fields are used as recorded for the exploratory summaries and model pipeline. Unknown is retained as a category rather than treated as missing. The presence of financial or demographic attributes does not establish that they are appropriate for deployment; purpose limitation and subgroup checks are needed.

2.3 DATA DICTIONARY: CAMPAIGN HISTORY AND TARGET

Table 3 — Campaign, contact-history and target fields.

Field	Type	Description / modeling note
contact	Category	Recorded communication channel.
day	Integer	Day of month for the recorded contact.
month	Category	Month label for the recorded contact.
duration	Integer	Call duration in seconds; excluded from pre-call prediction.
campaign	Integer	Current-campaign contact count recorded for the row.
pdays	Integer	Days since previous contact; −1 used as a sentinel in this file.
previous	Integer	Number of earlier contacts.
poutcome	Category	Previous campaign result: failure, success, other or unknown.
deposit	Target	Recorded outcome: yes/no; yes maps to positive class.

The target is mapped to 1 for yes and 0 for no. It is never supplied as a predictor. The -1 pdays value is retained as an observed numeric sentinel in this model; its precise interpretation should be verified against the original dataset documentation.

2.4 DATA QUALITY AND LIMITATIONS

Table 4 — Initial data-quality checks.

Check	Observed result	Interpretation
Rows × columns	11,162 × 17	Supplied file as received.
Blank cells	0	No null cells detected by the initial audit.
Exact duplicate rows	0	Does not prove distinct people without an ID.
Target labels	yes / no	Both labels are present.
Class mix	5,289 yes; 5,873 no	47.38% positive; 52.62% negative.
Age range	18–95	Review source values if provenance becomes available.
Balance range	−6,847–81,204	Skew and extreme values; unit not verified.
pdays sentinel	−1 in 74.57%	Retained; interpretation requires source confirmation.

No rows were dropped in the base quality audit, and no outlier was automatically deleted. A row with no missing cells can still have a nonstandard category or ambiguous sentinel. The no-duplicate result applies only to exact row equality and cannot rule out multiple records for the same individual.

2.5 DATA CLEANING AND ANALYSIS ASSUMPTIONS

Column names are checked against the expected 17-field schema, string labels are trimmed, and the target is normalized to lowercase yes/no values. Numeric fields are read as numbers and checked for missingness and plausible ranges. Since the supplied file has no blank cells, no rows are removed for null values in the initial analysis.

For model execution, imputation remains inside a scikit-learn pipeline so the same workflow can handle future incomplete records. Numeric columns use the training-fold median and categorical columns use the training-fold most frequent value. Unknown categorical levels are ignored safely by one-hot encoding.

Balance values are retained, including negative amounts, because they are not automatically invalid. Charts may use a signed logarithmic transformation to display a wide range, but the model uses standard scaling on the original numeric field. No target oversampling is applied.

3. EDA PROCESS: PROJECT WORKFLOW

The analysis proceeds from schema and label checks to descriptive summaries, visual comparisons, preprocessing, model fitting and held-out evaluation. The data flow below distinguishes descriptive exploration from the training and testing stage.

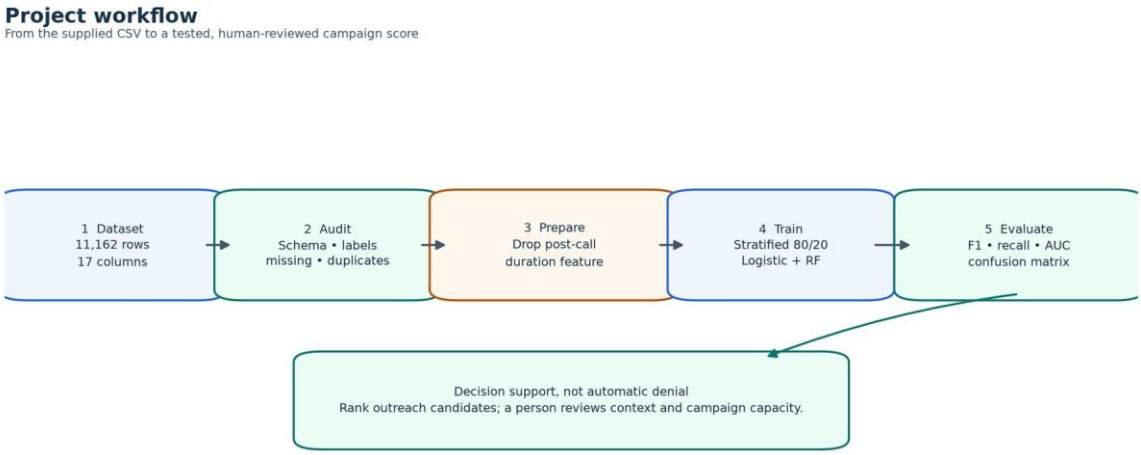


Figure 1 — Data-to-evaluation workflow.

- Audit records and document target proportions.
- Explore numerical distributions and categorical outcome rates.
- Define which inputs are available before a call.
- Fit preprocessing on the training fold only.
- Compare a transparent baseline with a nonlinear classifier.
- Review errors and communicate uncertainty and limitations.

3.1 DESCRIPTIVE STATISTICS AND TARGET PREVALENCE

Descriptive statistics summarize the scale and spread of numeric fields. Mean and standard deviation describe arithmetic variation; median and interquartile range are more robust when distributions are skewed. Categorical counts show how much evidence is available for each subgroup.

Table 5 — Selected numeric summaries by recorded outcome.

Measure	deposit=no	deposit=yes	Interpretation
Records	5,873	5,289	Both classes have substantial representation.
Age mean / median	40.84 / 39	41.67 / 38	Central age differs little.
Balance mean / median	1,280 / 414	1,804 / 733	Positive group has higher sample balance.
Duration median	163 s	426 s	Descriptive only; duration excluded from pre-call model.
Campaign mean / median	2.84 / 2	2.14 / 2	Repeat contacts differ in this sample.

The target distribution is 5,289 yes (47.38%) versus 5,873 no (52.62%). A majority-class model therefore has 52.62% accuracy but identifies no positive records; this is why accuracy must be interpreted with class-specific metrics.

EDA OBJECTIVE 1: TARGET CLASS DISTRIBUTION

Figure 2 presents the corresponding result generated from the supplied dataset.

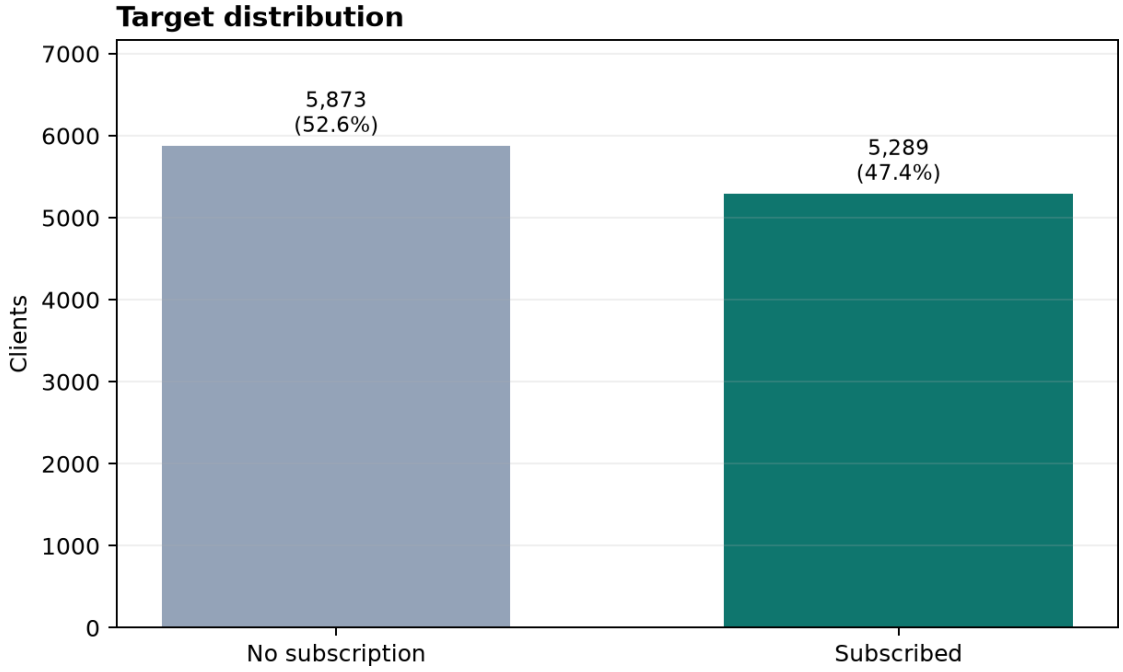


Figure 2 — Counts and proportions of the yes/no outcome.

The supplied sample is moderately balanced. This supports using a stratified train/test split and comparing precision and recall, not accuracy alone. The observed prevalence is specific to this CSV and should not be assumed to equal the rate in a future campaign.

EDA OBJECTIVE 2: AGE PROFILE AND SUBSCRIPTION

Introduction

Age can describe the composition of the records and reveal whether the positive outcome is concentrated in different age bands. Because age is personal information, subgroup patterns are presented descriptively and not as a basis for exclusion.

General description

Age values range from 18 to 95 years. The mean is 41.23, median is 39, and the interquartile range is 32–49. Mean age is 41.67 in yes records and 40.84 in no records; medians are 38 and 39.

Specific requirements, functions and formula

Age bands use intervals 18–29, 30–39, 40–49, 50–59, 60–69 and 70+. For each band, calculate n_g and the positive share r_g = count(deposit=yes in g) / n_g. Display the sample size beside the rate.

$$r_g = n(\text{deposit} = \text{yes}, \text{age} \in g) / n(\text{age} \in g)$$

Analysis results

Observed yes rates are 59.8% for ages 18–29 (1,551 rows), 44.3% for 30–39 (4,318), 40.4% for 40–49 (2,628), 41.6% for 50–59 (1,885), 75.4% for 60–69 (484), and 79.4% for 70+ (296). Older bands contain fewer records and require caution.

EDA OBJECTIVE 2: AGE-BAND VISUALIZATION

Figure 3 presents the corresponding result generated from the supplied dataset.

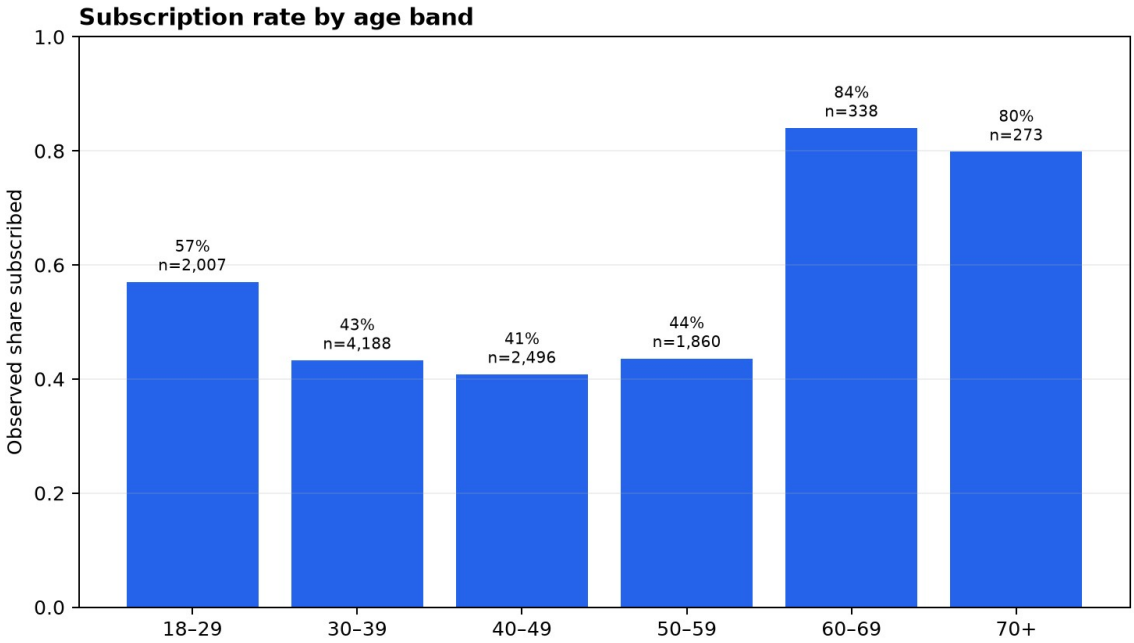


Figure 3 — Observed subscription rate by age band.

Rates in smaller bands can fluctuate more. The difference between age groups is associational, not causal; the two-sided Mann–Whitney test for age distributions by target gives p≈0.199. The project does not use this chart to define customer eligibility.

EDA OBJECTIVE 3: ACCOUNT BALANCE DISTRIBUTION

Introduction

Balance is a highly skewed financial field. Examining it by target helps identify scale, spread, and potential outlier sensitivity before choosing preprocessing.

General description

Balance has mean 1,528.54 and median 550; the interquartile range is 122–1,708. The range is −6,847 to 81,204. There are 688 negative and 774 zero values. For yes records the mean/median are 1,804.27/733; for no records they are 1,280.23/414.

Specific requirements, functions and formula

Use a signed logarithmic display transform $z=\text{sign}(x)\cdot\log(1+|x|)$ for the histogram only. This preserves sign while compressing extreme magnitudes. The classifier receives the original balance values and applies training-fold standardization.

$$\mathbf{z = sign(balance) \times ln(1 + |balance|)}$$

Analysis results

The positive group has higher mean and median balance in this sample, but the distribution is skewed and includes large values. The chart is a descriptive comparison and does not imply balance causes subscription.

EDA OBJECTIVE 3: BALANCE VISUALIZATION

Figure 4 presents the corresponding result generated from the supplied dataset.

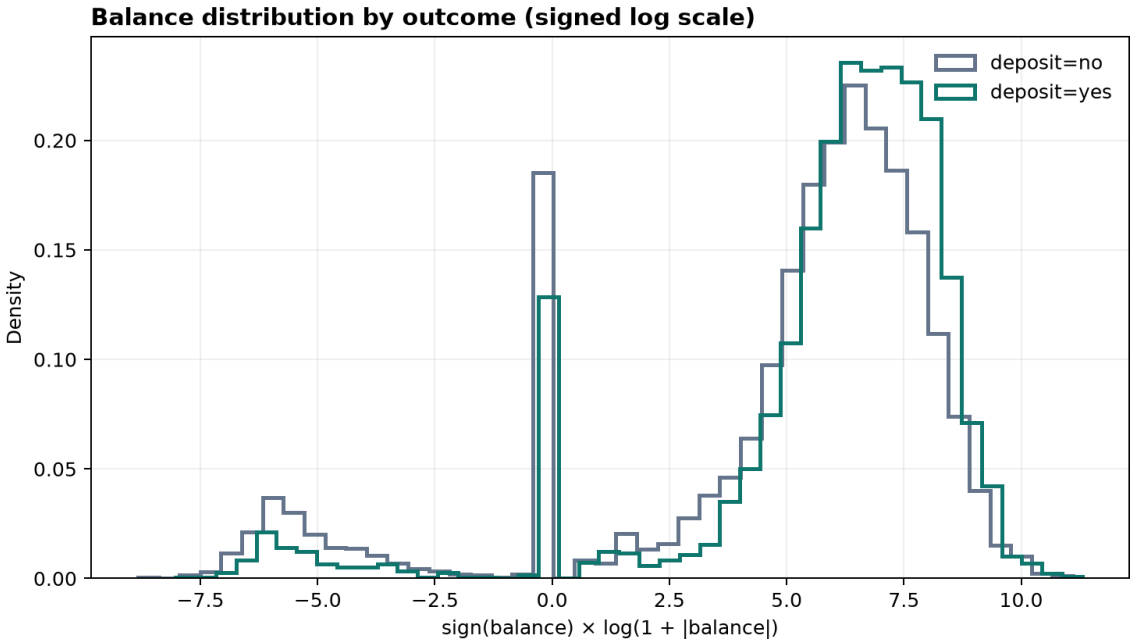


Figure 4 — Signed-log distribution of balance by target label.

The signed-log scale is a visualization aid, not a redefinition of the underlying unit. The report keeps original-scale summary statistics alongside the chart so that the magnitude and data limitations remain visible.

EDA OBJECTIVE 4: LOAN AND CONTACT ATTRIBUTES

Introduction

Housing-loan, personal-loan and contact-channel categories can highlight differences in the observed response mix. This analysis summarizes proportions rather than comparing customer suitability.

General description

Housing=no records have a 57.0% yes rate versus 36.6% for housing=yes. Personal-loan=no records have a 49.5% yes rate versus 33.2% for loan=yes. Contact rates are 54.3% for cellular, 50.4% for telephone and 22.6% for unknown.

Specific requirements, functions and formula

For category c, compute r_c = n(yes,c)/n(c), and report n(c) next to each proportion. The chi-square test checks independence between the category and target; Cramér's V gives a scale-adjusted association size.

$$r_c = n(\text{deposit} = \text{yes} \mid \text{category} = c) \,/\, n(\text{category} = c)$$

Analysis results

Housing has Cramér's V=0.204; contact has V=0.257; personal loan has V≈0.110. These are sample associations, can be confounded by campaign selection, and do not support causal or individual-level claims.

EDA OBJECTIVE 4: CATEGORY-RATE VISUALIZATION

Figure 5 presents the corresponding result generated from the supplied dataset.

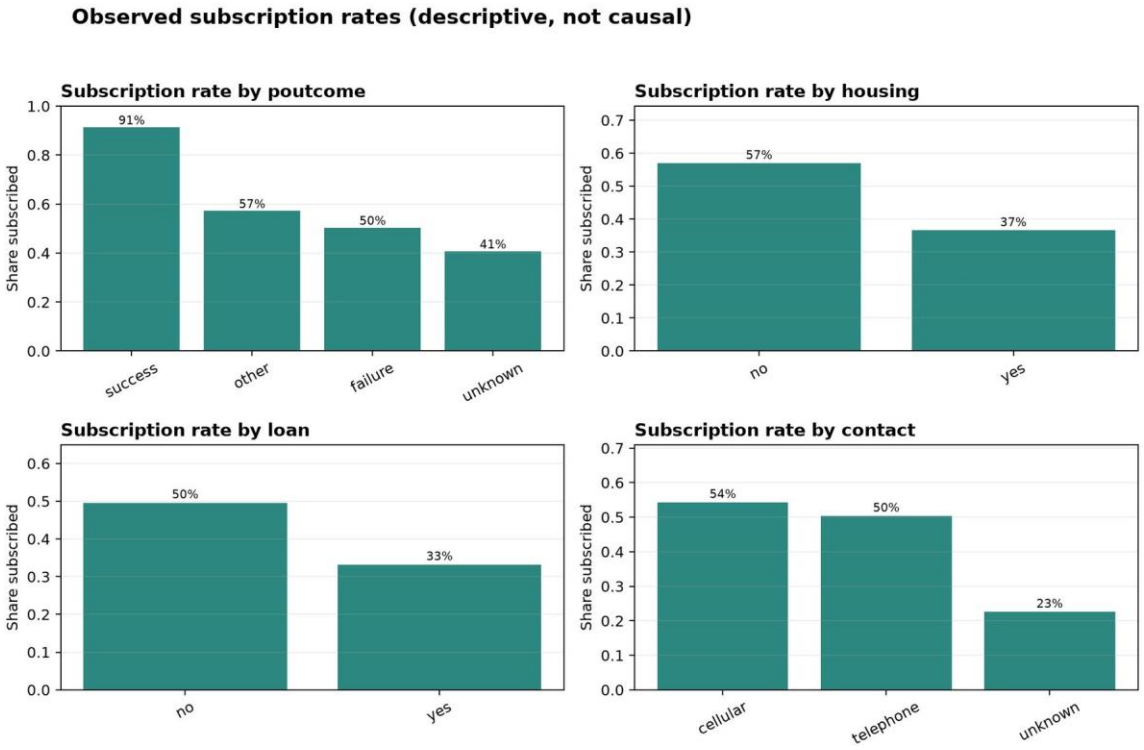


Figure 5 — Observed yes rates for selected customer and campaign categories.

The four panels show previous outcome, housing loan, personal loan and contact type. Subgroup counts differ substantially; the chart labels group rates for comparison but does not adjust for other features.

EDA OBJECTIVE 5: PREVIOUS CAMPAIGN OUTCOME

Introduction

Past campaign response may provide useful historical context for a new outreach decision. The distribution is examined together with the positive class rate and group size.

General description

The yes rate is 91.3% for poutcome=success (1,071 rows), 50.3% for failure (1,228), 57.2% for other (537), and 40.7% for unknown (8,326). Unknown is the largest group and dominates the total count.

Specific requirements, functions and formula

Group the records by poutcome, count the target labels and calculate r_c for each level. A chi-square test gives Cramér's V=0.300 with p<0.001 in this sample.

$$\text{Cramér's V} = \text{sqrt}(\chi^2 / (N \times \min(r-1, c-1)))$$

Analysis results

Prior success is strongly associated with the outcome in the observed dataset. It may encode prior selection and campaign history; the association is not evidence that repeating the same contact strategy causes a subscription.

EDA OBJECTIVE 5: PRIOR-OUTCOME VISUALIZATION

Figure 6 presents the corresponding result generated from the supplied dataset.

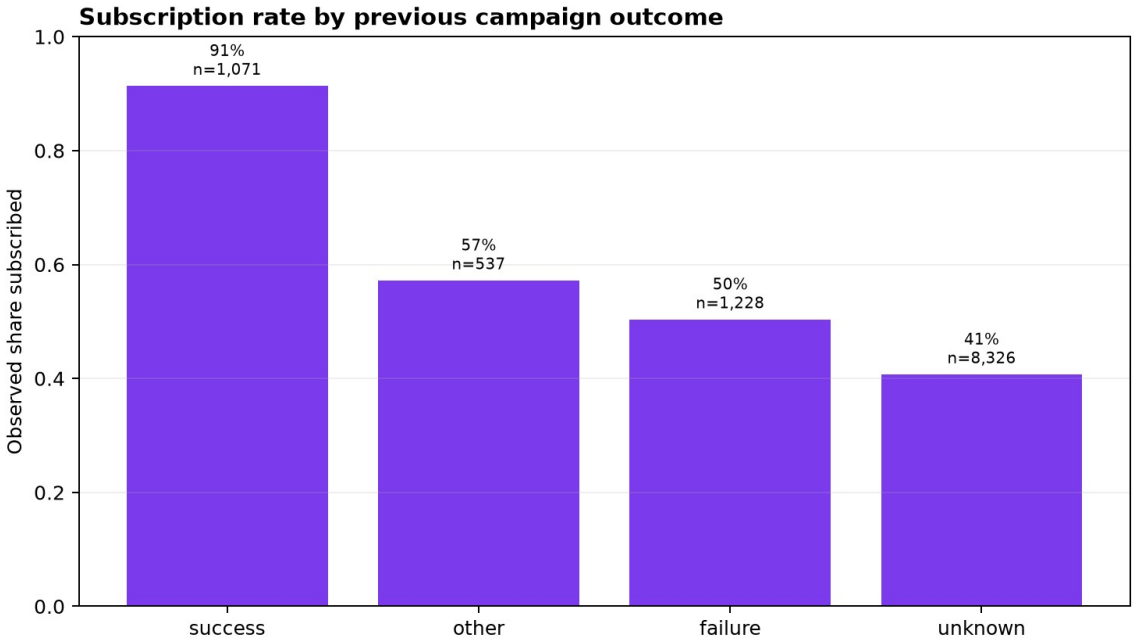


Figure 6 — Observed subscription rate by previous campaign result.

The success category has a high observed rate but represents a minority of the full sample. The largest group is unknown. Any operational model should monitor the quality and availability of prior-outcome data over time.

EDA OBJECTIVE 6: CURRENT-CAMPAIGN CONTACT INTENSITY

Introduction

Contact frequency can be related to both response and campaign effort. A grouped summary can show whether repeated contact counts are associated with different recorded response rates.

General description

The campaign variable has median 2 and mean 2.51. Contact count bands are 1, 2, 3, 4–5, 6–10 and 11+. The observed rates are 53.4%, 46.3%, 46.8%, 39.7%, 31.4% and 22.4%, respectively.

Specific requirements, functions and formula

For each contact-count band, report the proportion of yes labels and group count. The band edges are selected for readability; they are not optimized thresholds.

$$\text{Observed rate in band } b = n(\text{yes},b) / n(b)$$

Analysis results

The declining pattern at high counts may reflect response history and selection into repeated calls. It does not show that additional calls reduce subscription probability; a randomized experiment would be needed to measure causal contact effects.

EDA OBJECTIVE 6: CONTACT-COUNT VISUALIZATION

Figure 7 presents the corresponding result generated from the supplied dataset.

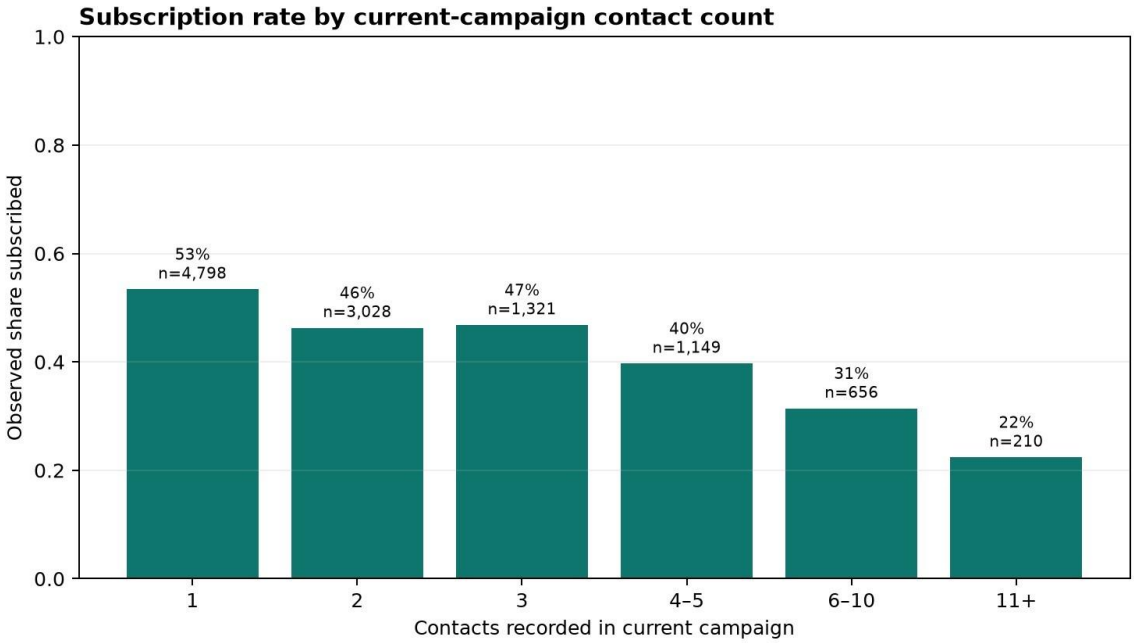


Figure 7 — Observed yes rate by contact count in the current campaign.

The smallest/highest-count categories have fewer observations. The rate chart includes n labels to make this limitation visible; a future analysis should report confidence intervals.

EDA OBJECTIVE 7: CAMPAIGN MONTH AND TIMING

Introduction

Campaign month may capture scheduling, operational or sample composition differences. Monthly comparisons help describe when the supplied campaign records were associated with more yes labels.

General description

Rates range from 32.8% in May (2,824 rows) to 90.9% in December (110 rows). March is 89.9% (276 rows), September 84.3% (319), and October 82.4% (392). Month has Cramér's V=0.306 in the exploratory chi-square test.

Specific requirements, functions and formula

Order months from January through December, calculate r_m and n_m for each month, and display both. The chi-square analysis compares the full target distribution across month categories.

r_m = n(deposit = yes in month m) / n(month m)

Analysis results

High rates in smaller months can be unstable. Calendar patterns may reflect campaign scheduling or sampling rather than a durable seasonal response effect.

EDA OBJECTIVE 7: MONTHLY VISUALIZATION

Figure 8 presents the corresponding result generated from the supplied dataset.

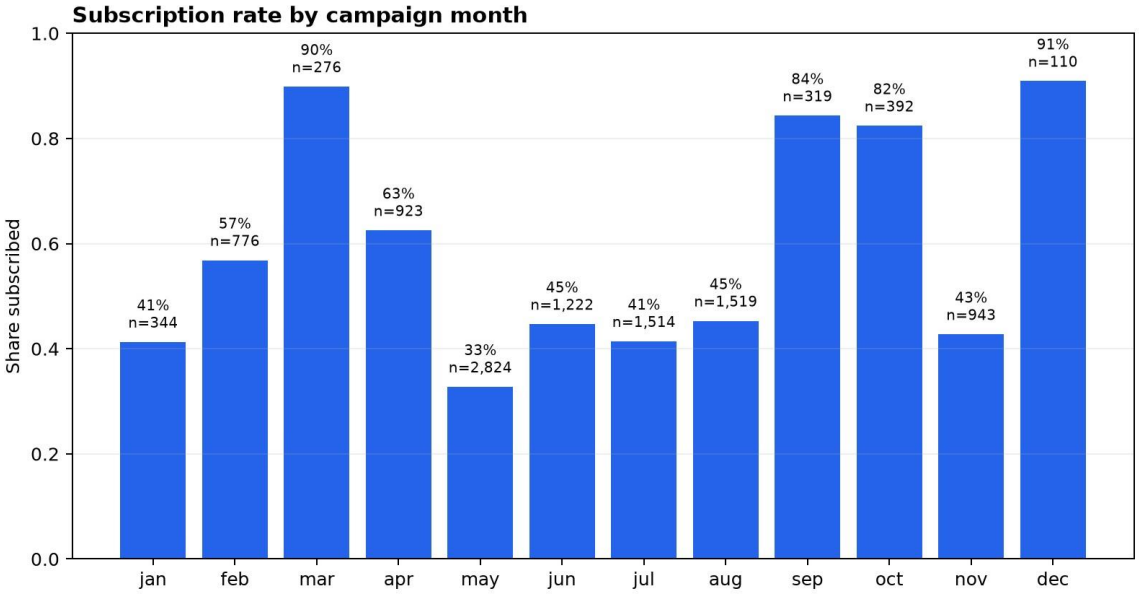


Figure 8 — Observed subscription rate by month with subgroup counts.

May is the largest monthly group and has a comparatively low observed yes rate. Several high-rate months have fewer than 400 records. Results should be rechecked on a time-based evaluation set before making campaign-calendar recommendations.

3.8 STATISTICAL ASSOCIATION CHECKS

Pearson chi-square tests compare target-label distributions across categorical levels. Cramér's V is reported as an effect-size summary. Two-sided Mann–Whitney tests compare numeric ranks by target, which is suitable for skewed balance and campaign counts. Tests are exploratory, share the same dataset, and are not adjusted for multiple comparisons.

Table 6 — Selected exploratory tests.

Field	Test / effect	Result	Caution
Contact	x^2; V=0.257	p<0.001	Association only.
Month	x^2; V=0.306	p<0.001	Small month groups vary in size.
Poutcome	x^2; V=0.300	p<0.001	History may reflect selection.
Housing	x^2; V=0.204	p<0.001	No causal inference.
Job	x^2; V=0.184	p<0.001	Many categories; no multiple-test correction.
Age	Mann–Whitney	p≈0.199	No clear rank-distribution difference.
Balance	Mann–Whitney	p<0.001	Skew and extremes matter.

A small p-value does not measure business importance and cannot replace held-out predictive evaluation. Association strength, sample size, selection bias and practical impact all require separate consideration.

4. PREDICTION-TIME BOUNDARY AND LEAKAGE CONTROL

The model is intended to rank records before a new outreach call. Call duration belongs to the period after the contact begins and can strongly reveal the outcome. Using it for pre-call scoring would allow information unavailable at prediction time to enter the model.

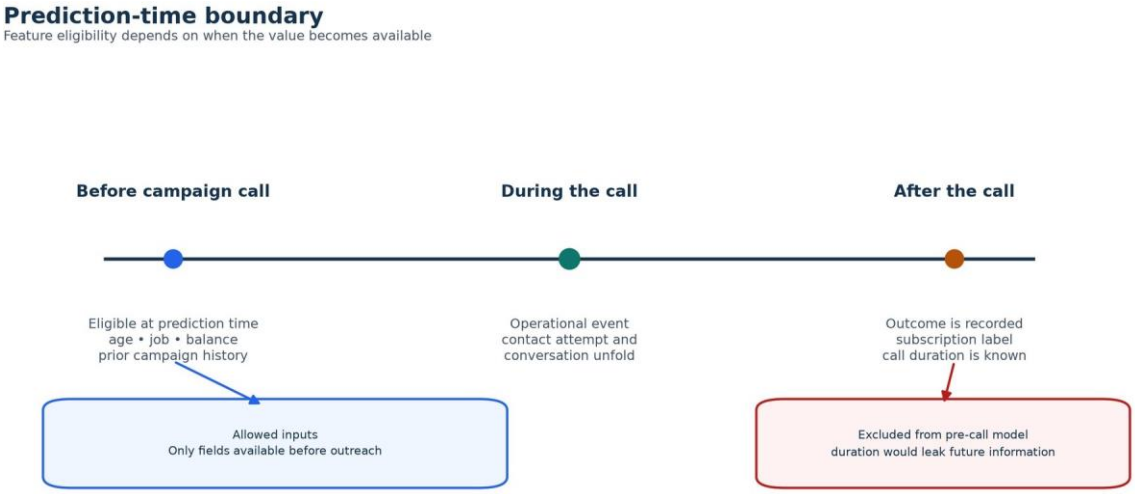


Figure 9 — Feature availability before, during and after a call.

Duration is excluded from all model inputs, feature importance and test metrics. Month/day are retained for this offline study only under the assumption that the scheduled campaign timing is known before scoring. Reassess those fields if the deployment prediction time changes.

4.1 FEATURE ENGINEERING AND VALIDATION SPLIT

The target is deposit, encoded as yes=1 and no=0. The feature frame removes deposit and duration. Numeric fields are age, balance, day, campaign, pdays and previous; categorical fields include the remaining eligible profile and campaign categories.

The dataset is partitioned into 80% training and 20% testing with stratification and random_state=42. The training set contains 8,929 records and the test set contains 2,233. The test set contains 1,175 no labels and 1,058 yes labels.

Median imputation and standard scaling are fitted to numeric training values; most-frequent imputation and one-hot encoding are fitted to categorical training values. The resulting transformations are applied to the test set without refitting. This is a random holdout, not a chronological validation.

4.2 MODEL 1: LOGISTIC REGRESSION

Logistic regression provides a linear baseline for the binary subscription outcome. It models the conditional probability of yes using a logistic link applied to an encoded linear score.

$$P(Y=1|X) = 1 / (1 + \exp[-(\beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p)])$$

The implementation uses maximum iterations of 1,000 and random_state=42. Numeric scaling supports optimization; one-hot encoding represents categories. The default probability threshold of 0.50 is used for class predictions to keep the comparison reproducible.

Coefficients in a fitted logistic model would describe conditional associations given the encoded inputs, not causal effects. This project does not tune thresholds or interpret coefficients as customer motivations.

4.3 MODEL 2: RANDOM FOREST AND PREPROCESSING PIPELINE

Random forest combines decision trees fitted on bootstrap samples with randomized feature selection. The chosen configuration uses 250 estimators, maximum depth 18, minimum leaf size 3, balanced-subsample class weights, and random_state=42.

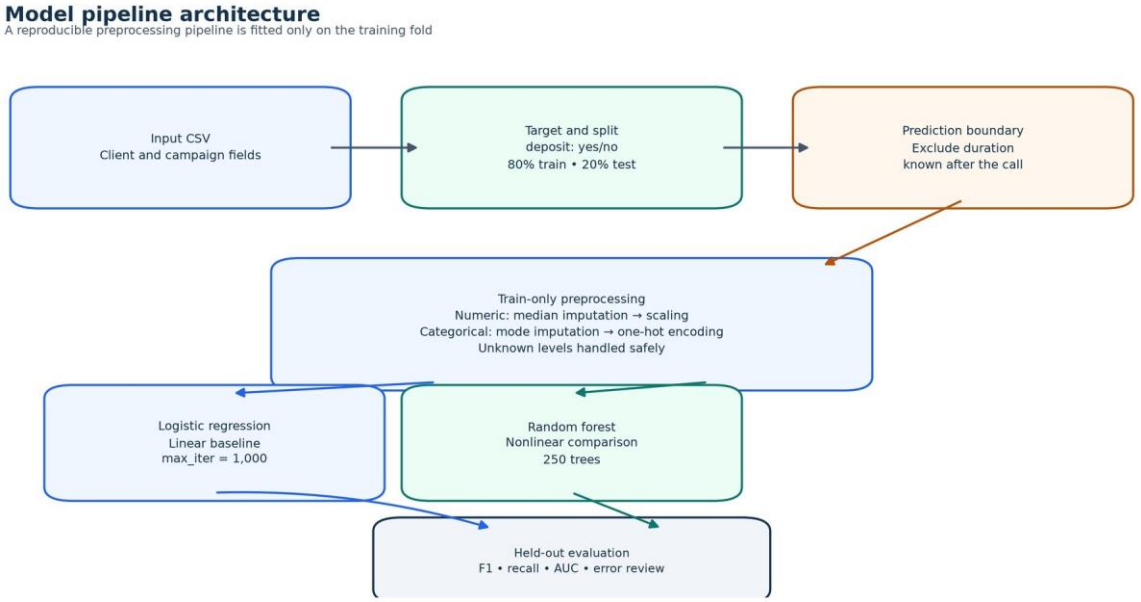


Figure 10 — Preprocessing branches and model-comparison flow.

The configuration is a bounded comparison, not an exhaustive hyperparameter search. Class weighting can affect probability calibration and the precision/recall trade-off. Future evaluation should use cross-validation or a validation fold for tuning and preserve an untouched time-based test set.

4.4 EVALUATION METRICS AND FORMULAS

The positive class is deposit=yes. Precision, recall, F1, accuracy and balanced accuracy use class predictions at threshold 0.50. ROC AUC uses the predicted probability and evaluates ranking across thresholds.

Metric	Formula	Interpretation
Accuracy	(TP+TN)/N	Overall share correctly classified.
Precision	TP/(TP+FP)	Share of predicted yes that are yes.
Recall	TP/(TP+FN)	Share of actual yes identified.
F1	2×P×R/(P+R)	Harmonic balance of precision and recall.
Balanced accuracy	(TPR+TNR)/2	Average class-specific recall.
ROC AUC	Area under TPR–FPR curve	Ranking ability, not calibration or campaign lift.

The business costs of false positives and false negatives are not specified. Therefore, the 0.50 threshold is a reporting convention rather than a recommended operating threshold.

4.5 MODEL PERFORMANCE COMPARISON

All classifiers are evaluated on the same stratified test partition. The majority baseline predicts the most frequent training class for every row, establishing a simple reference point.

Table 7 — Held-out test metrics (positive class: yes).

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Most-frequent baseline	0.526	0.500	0.000	0.000	0.000	0.500
Logistic regression	0.695	0.689	0.733	0.562	0.636	0.758
Random forest	0.730	0.725	0.760	0.630	0.689	0.787

Random forest is strongest on the reported measures for this split. It improves F1 by about 0.053 over logistic regression and ROC AUC by about 0.029. It remains imperfect: 391 positive records are missed at threshold 0.50.

MODEL METRIC VISUALIZATION

Figure 11 presents the corresponding result generated from the supplied dataset.

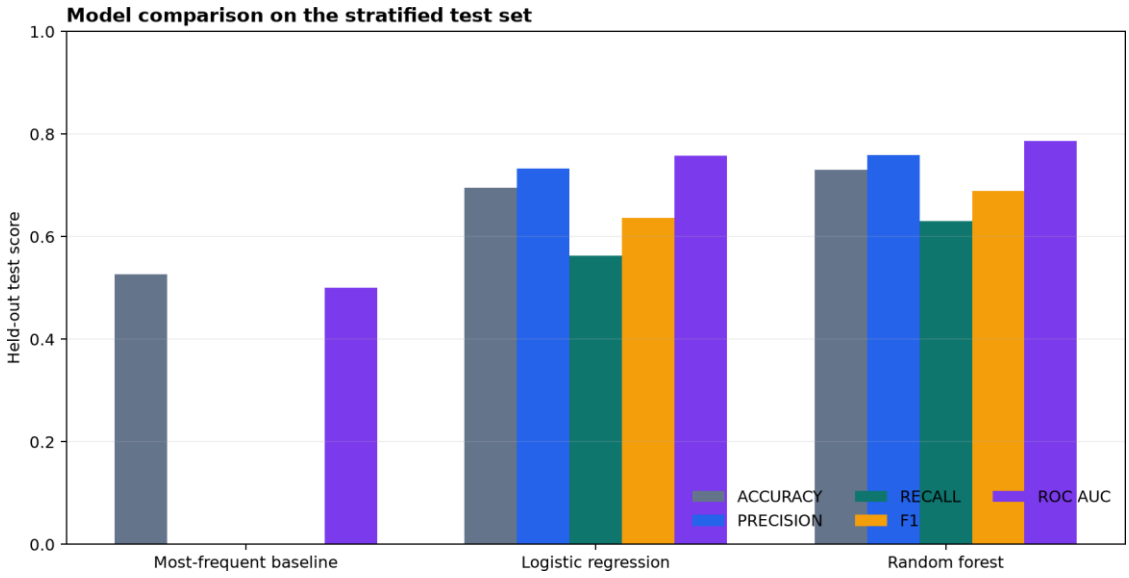


Figure 11 — Comparison of test-set performance measures.

The plot compares accuracy, precision, recall, F1 and ROC AUC. The majority baseline illustrates why accuracy without positive class performance can be misleading.

4.6 ERROR ANALYSIS: RANDOM FOREST

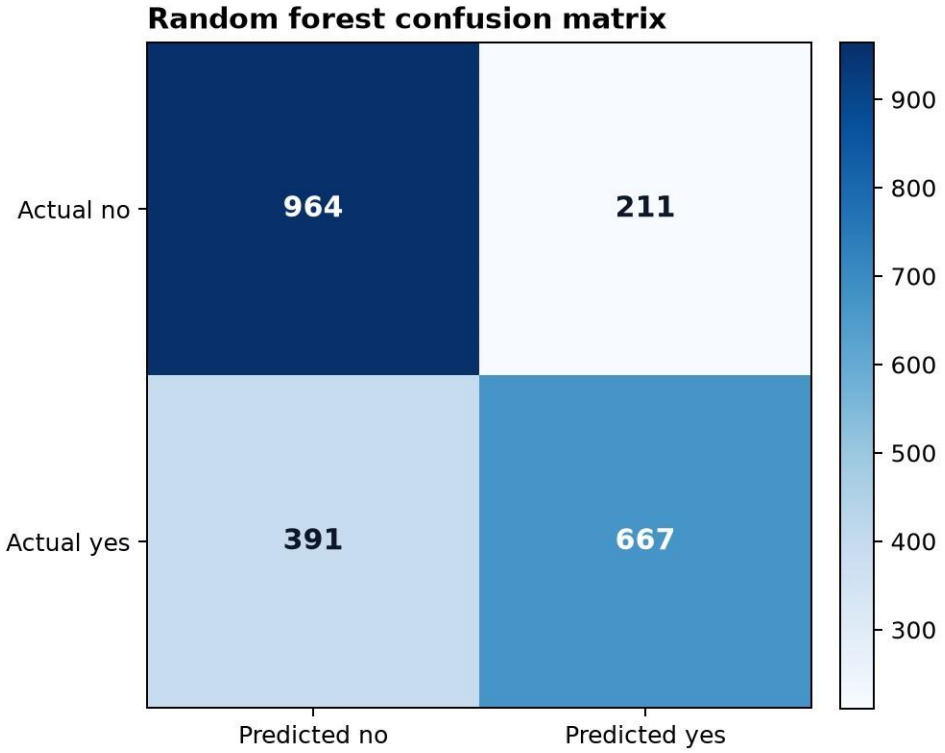


Figure 12 — Actual labels are rows and predicted labels are columns.

Table 8 — Random-forest confusion-matrix counts.

	Predicted no	Predicted yes	Actual total
Actual no	964 (TN)	211 (FP)	1,175
Actual yes	391 (FN)	667 (TP)	1,058
Predicted total	1,355	878	2,233

The model produces 211 false positives and 391 false negatives. Precision is 0.760 and recall is 0.630. Whether this trade-off is acceptable depends on outreach capacity, contact cost and customer experience; those costs were not measured in this study.

4.7 ROC CURVE ANALYSIS

Figure 13 presents the corresponding result generated from the supplied dataset.

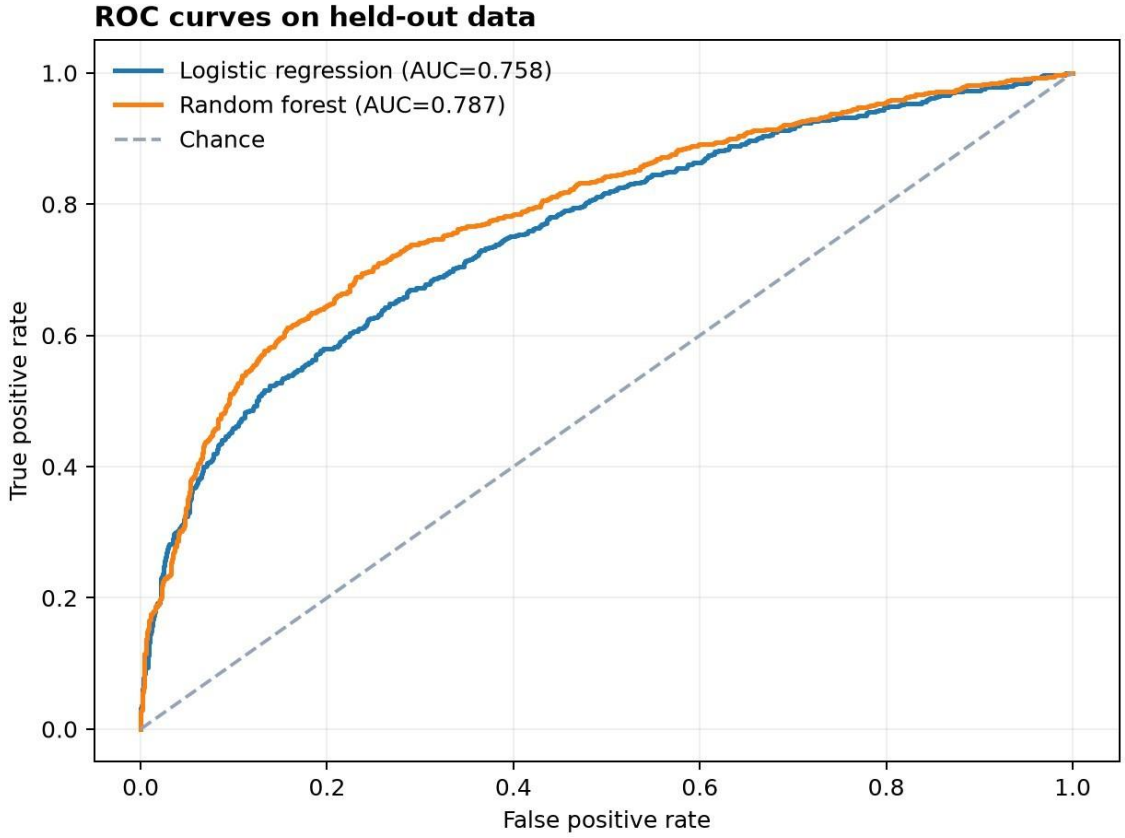


Figure 13 — ROC curves for learned models on the same test fold.

Random forest has ROC AUC 0.787 and logistic regression 0.758; the majority baseline is 0.500. AUC summarizes ranking across thresholds; it does not mean that 78.7% of individual predictions are correct and does not establish campaign lift.

4.8 FEATURE IMPORTANCE AND INTERPRETATION

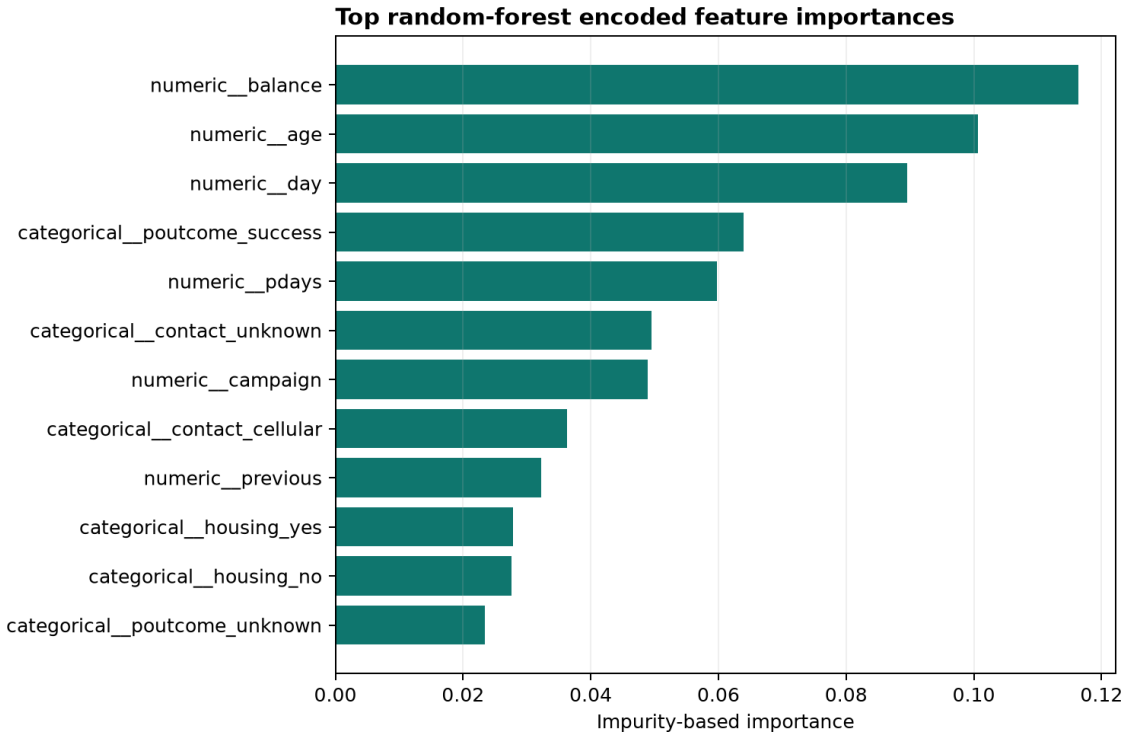


Figure 14 — Top encoded-feature impurity importances.

The largest individual encoded importances include balance (0.116), age (0.101), day (0.090), poutcome_success (0.064), pdays (0.060), contact_unknown (0.049), and campaign (0.049). These are model-specific split importances, not the direction of a relationship, causal effects, or a stable ranking across samples.

Categorical levels divide one source feature across multiple indicators, and correlated inputs may share importance. A future study should check stability with permutation importance on a validation set and document explanations carefully. Duration is absent because it was excluded before training.

5. SOCIETAL VALUE AND RESPONSIBLE USE

The intended benefit is a possible improvement in campaign planning: staff may prioritize reviewing records that have higher estimated subscription likelihood. This can only be described as a potential benefit until an intervention is tested against a control group.

- Use scores only as optional outreach prioritization, not to deny products or alter financial terms.
- Respect consent, contact preferences, opt-outs and campaign-frequency limits.
- Keep a human review step and document the scoring version and threshold.
- Minimize personal data and evaluate group-level error rates with appropriate safeguards.
- Monitor drift, calibration, complaints and campaign outcomes; pause use if safeguards fail.

This project includes no legal compliance assessment, fairness certification, customer research or production security review. Historical data may encode unequal prior selection even when a sensitive field is omitted.

6. LIMITATIONS, VALIDITY AND FUTURE IMPROVEMENTS

Limitations

- The exact original CSV source, sampling process, licence and time span are unverified.
- There is no customer ID, so repeat individuals and group leakage cannot be ruled out.
- Random splitting may mix records from different campaign periods and does not test future-period generalization.
- Only one holdout split and two model families are reported; no tuning or calibration study was conducted.
- The project does not test whether model-ranked calls cause incremental subscriptions.
- No fairness, privacy, security or production-monitoring assessment was completed.

Validity checks for a next study

A future version should verify the dataset source, use a time-ordered split, tune models on validation data, assess probability calibration, report precision-recall and top-k yield, and check error rates across appropriate subgroups. A prospective randomized campaign is needed to measure incremental benefit.

6.1 FUTURE SCOPE

- Reconcile the uploaded file with a verifiable publisher and document its licence.
- Use the campaign date or period to build a forward-chaining evaluation.
- Compare logistic regression, calibrated tree ensembles and other appropriate baselines.
- Choose thresholds from documented contact-capacity and false-positive/false-negative costs.
- Evaluate calibration, confidence intervals, subgroup performance and model stability.
- Run a controlled outreach experiment to measure incremental subscriptions and customer outcomes.
- Develop a small, access-controlled decision-support interface only after governance review.

These extensions would turn a reproducible classroom comparison into a stronger evidence base. None should be presented as completed until the corresponding analyses are run.

7. CONCLUSION

The supplied bank-marketing CSV contains 11,162 rows and 17 fields, with 47.38% yes and 52.62% no outcomes. The initial audit found no blank cells or exact duplicate records. Descriptive analysis shows variation in observed outcomes by prior campaign result, contact type, month, loan status, age band and current-campaign contact count.

With duration excluded to maintain a pre-call prediction boundary, random forest outperformed logistic regression on the reported metrics for one stratified test split. It achieved 73.04% accuracy, 0.689 F1 and 0.787 ROC AUC. At threshold 0.50 it still missed 391 of 1,058 positive test records.

The result is an academic decision-support prototype, not a production guarantee. Dataset source verification, time-based validation, threshold analysis, subgroup review, customer safeguards and experimental evidence are needed before practical use. The student should complete all blank details and links and verify every result against the separately supplied code before submission.

REFERENCES

1. Breiman, L. (2001). Random Forests. Machine Learning, 45, 5–32. https://doi.org/10.1023/A:1010933404324

2. Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems, 62, 22–31. https://doi.org/10.1016/j.dss.2014.03.001

3. UCI Machine Learning Repository. (2012). Bank Marketing, Dataset 222. https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing. Consulted for task context and dataset-version comparison only; exact identity of the supplied CSV is not established.

4. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

5. Scikit-learn developers. User Guide: preprocessing, linear models, ensemble methods and model evaluation. https://scikit-learn.org/stable/user_guide.html

6. pandas development team. pandas documentation. https://pandas.pydata.org/docs/

7. Matplotlib development team. Matplotlib documentation. https://matplotlib.org/stable/

8. Project dataset: bank_1793283222379.csv, supplied for this analysis. Original source and licence: [VERIFY AND ADD BEFORE SUBMISSION].

APPENDIX A: REPRODUCIBILITY AND PROJECT LINKS

The model implementation and diagram source are supplied separately from this report, as requested. The modeling run uses an 80/20 stratified split, random_state=42, median numeric imputation, standard scaling, most-frequent categorical imputation and one-hot encoding. Logistic regression uses max_iter=1000. Random forest uses 250 trees, maximum depth 18, minimum leaf size 3 and balanced-subsample class weights.

To reproduce final values, rerun the supplied scripts with the exact CSV used for submission and confirm the output metrics and figures. If the file, package versions, features or settings change, update this report before submission.

Complete the following links before submitting.

Required item	Space for URL
Project / demonstration link	[Paste project URL here]
Dataset source link	[Paste verified original dataset URL here]
LinkedIn post / profile	[Paste LinkedIn URL here]
GitHub repository	[Paste GitHub URL here]

Student and faculty particulars

Particular	Complete before submission
Student name and registration	[_____]
Programme, section and course code	[_____]
Supervisor name, U.Id and designation	[_____]
School and project semester	[_____]

Final check: verify source provenance and licence, confirm that every reported result matches the final run, update the contents after edits, and obtain supervisor review.